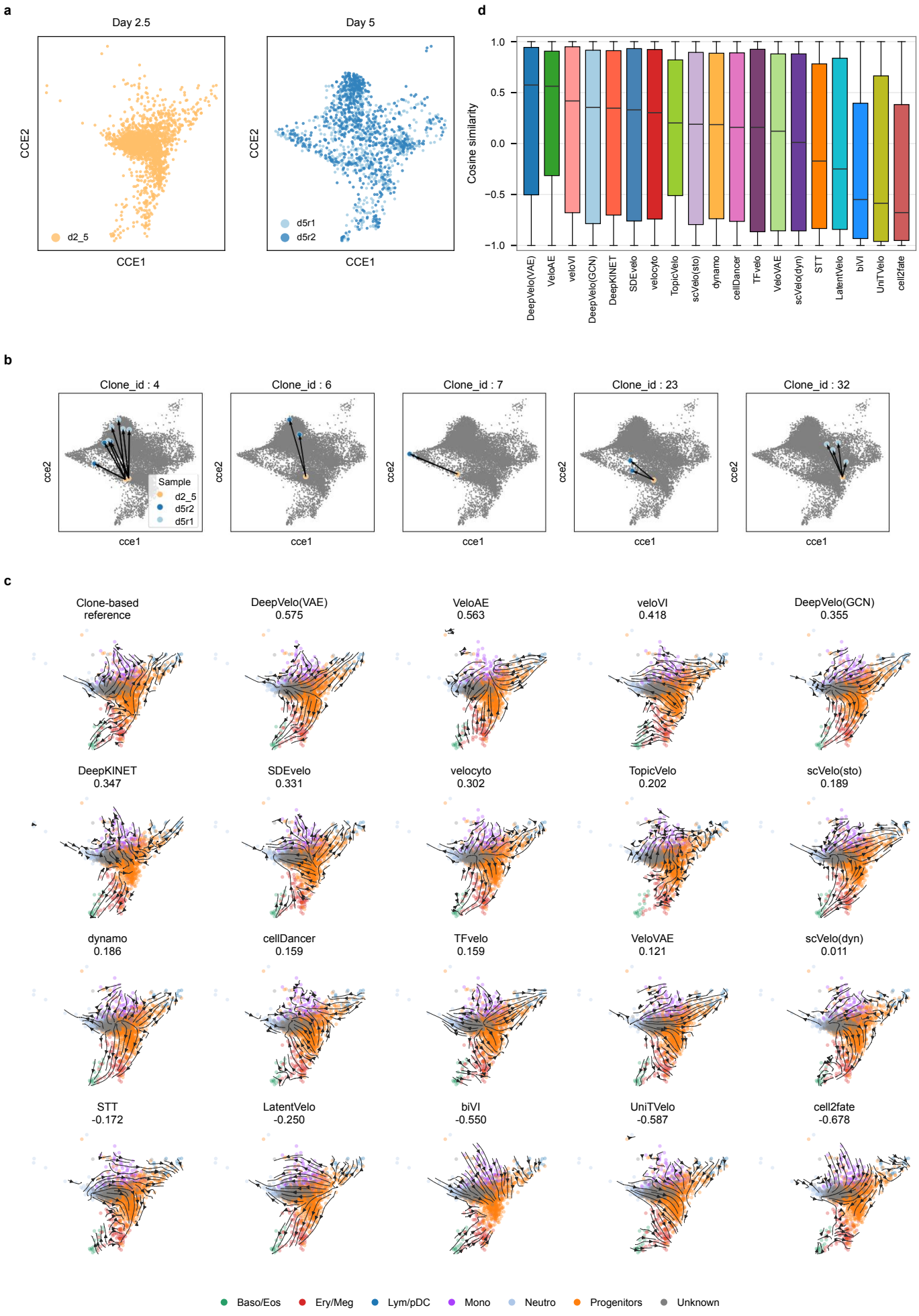

a
Day 2.5
Day 5
CCE2
CCE1
d2_5
d5r1
d5r2
d
Cosine similarity
1.0
0.5
0.0
−0.5
−1.0
DeepVelo(VAE)
VeloAE
veloVI
DeepVelo(GCN)
DeepKINET
SDEvelo
velocyto
TopicVelo
scVelo(sto)
dynamo
cellDancer
TFvelo
VeloVAE
scVelo(dyn)
STT
LatentVelo
biVI
UniTVelo
cell2fate
b
Clone_id : 4
Clone_id : 6
Clone_id : 7
Clone_id : 23
Clone_id : 32
cce2
cce1
Sample
d2_5
d5r2
d5r1
c
Clone-based reference
DeepVelo(VAE) 0.575
VeloAE 0.563
veloVI 0.418
DeepVelo(GCN) 0.355
DeepKINET 0.347
SDEvelo 0.331
velocyto 0.302
TopicVelo 0.202
scVelo(sto) 0.189
dynamo 0.186
cellDancer 0.159
TFvelo 0.159
VeloVAE 0.121
scVelo(dyn) 0.011
STT -0.172
LatentVelo -0.250
biVI -0.550
UniTVelo -0.587
cell2fate -0.678
Baso/Eos
Ery/Meg
Lym/pDC
Mono
Neutro
Progenitors
Unknown

Extended Data Fig. 8 | Lineage-tracing reference: scLT-seq. a, CCE views of Day 2.5 (1,756 cells) and Day 5 (675 and 755 cells in two replicates). **b,** Clone-transition examples for clone IDs 4, 6, 7, 23 and 32. **c,** Clone-based reference streams and all 19 method configurations for 1,756 cells, colored by cell type. **d,** Per-cell cosine similarity to the clone-based reference. Numbers below method names in the stream panels are the median per-cell reference-direction cosines, shown to three decimal places. Boxes show the median and interquartile range, with 1.5-IQR whiskers and outliers; methods are ordered by median. All 19 configurations use barcode-aligned cells and the same reference coordinates. Cosines were calculated from the displayed per-cell directions. TopicVelo has 1,724 valid cosines; its 32 zero-direction cells remain undefined and are excluded from its boxplot. All other configurations have 1,756 valid cosines.